MAJOR PROJECT · PHASE-I PROGRESS REPORT – 1 · B.TECH COMPUTER SCIENCE & ENGINEERING · PROJECT TEAM ID CSE27-293

MFSTNet — Multimodal Spatio-Temporal Congestion Forecasting for Adaptive Traffic Signal Control

working name: cnn-vit-bilstm-cross-attention-traffic-system

Progress Report – 1 covers the problem, the literature and the research gap. It reports no measurement: none has been made yet, and a target stated as a result is the overclaim this project's reporting rules exist to prevent.

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Three questions, in the order they get asked

- **Is the problem real?**
  slides 3-6 · what a fixed-time signal cannot do, and why the stream makes it worse
- **Do we know the literature?**
  slides 7-12 · four settled bodies of work, 30 papers compared
- **Is anything left to do?**
  slides 13-15 · the gap, derived from the survey rather than asserted

WHAT THIS REPORT IS

- **Five sections, no more**
  the guideline prescribes exactly these
- **1 Introduction**
  nine named parts
- **2 Literature Survey**
  ten themed subsections
- **3 Literature Survey Table**
  30 papers, seven columns
- **4 Research Gap**
  five questions answered
- **References**
  53 entries, all cited

Every citation on every slide resolves to a record that was retrieved from a bibliographic index, not recalled.

A signal allocates green from a survey taken once

Fixed-time control computes cycle length and phase splits from a manual count, then holds them for years.

- **The plan is right on average**
  and therefore wrong most of the time — green is held on an empty approach while a saturated one waits
- **Re-timing needs a new survey**
  so plans age; the method itself dates from 1958 and is still the basis of Indian practice
- **The saturation flow it needs is not a constant**
  published values for one intersection differ about two-fold with the PCU convention alone

WHY CAMERA-ONLY

- **Cameras already exist**
  installed for enforcement, at most junctions
- **Loops and probes do not**
  and cost money the deployment does not have
- **A decision acts on the future**
  the next 30-90 s, not the present queue
- **So forecast, do not measure**
  one cycle ahead is the useful horizon

A camera that is already mounted, already powered and already pointed at the stop line is the one sensor that can be used without installing anything.

The stream is not the stream the method assumes

Heterogeneous, non-lane-disciplined, and Indian practice already says counting vehicles measures the wrong thing.

WHAT ARRIVES AT THE JUNCTION

- **Mixed modes on one approach**
  two-wheelers, auto-rickshaws, e-rickshaws, cars, buses, trucks
- **Different road space, different discharge**
  a bus and a two-wheeler are not one vehicle each
- **No lane discipline**
  queue geometry does not follow the markings on the road
- **Pedestrians and animals in the carriageway**
  on many of the roads this targets

- **So congestion is defined on occupancy**
  passenger-car-unit weighted, not a raw count — the Indian Highway Capacity Manual publishes the equivalences for this reason
- **And thresholds are calibrated per camera**
  a threshold correct at one viewpoint is arbitrary at another
- **And whole cameras are held out**
  a random split over overlapping windows lets a model recognise a camera it has already seen and call that generalisation

The problem statement

Every clause is a constraint, not a preference.

Given only the video of a single fixed camera already installed at a signalised urban intersection, predict the congestion level of each approach lane sixty seconds ahead — accurately enough for a controller to act on, under heterogeneous non-lane-disciplined traffic, without a raw frame ever leaving the device.

WHY EACH CLAUSE IS THERE

- **A single camera**
  several per junction is not deployable here
- **Per-lane output**
  a controller allocates green to an approach
- **Sixty seconds ahead**
  a forecast of the present is a measurement
- **No raw frames off-device**
  otherwise it is a surveillance system

SIX OBJECTIVES

- **O1 Build the labelled fixed-camera corpus**
- **O2 Fine-tune a detector for Indian classes**
- **O3 Train the multimodal forecaster**
- **O4 Show whether each component earns its place**
- **O5 Couple the forecast to a controller**
- **O6 Keep it reproducible and privacy-preserving**

Scope, and what is outside it

An unstated exclusion reads as an unmet expectation at evaluation time.

WITHIN SCOPE

- **Per-lane congestion forecast, 60 s ahead, one fixed camera**
- **A curated fixed-camera corpus of heterogeneous traffic**
- **A detector fine-tuned for eight Indian vehicle classes**
- **An ablation, and count-only baselines to beat**
- **An RL signal controller, evaluated in simulation**
- **An edge-to-server prototype that moves no raw frame**

OUTSIDE SCOPE

- **Field deployment on a live signal**
- **Multi-intersection coordination**
- **Re-identification, tracking, number-plate recognition**
- **Trajectory prediction for individual vehicles**
- **A production data platform**

Four settled bodies of work, and what each leaves

Organised by what a strand settles, not by date — the question is what is left, not what exists.

THE FOUR STRANDS

- **Congestion from camera imagery**
  settled: the signal is in the frame. Left: scene-level, present-tense
- **Forecasting on sensor graphs**
  settled: accurate on instrumented networks. Left: no loops here
- **CNN-transformer fusion with a gate**
  settled: published, some of it in 2019. Left: does it help this task
- **RL signal control**
  settled: benchmarked to city scale. Left: anticipatory state

- **We claim none of the four**
  each is cited as prior art; claiming one invites a citation we did not survey
- **What survives is the conjunction**
  per-lane, one phase ahead, one fixed camera, heterogeneous traffic, coupled to a controller

The task exists; the sensing assumption is the gap

CAMERA-BASED CONGESTION

- **Chakraborty et al. (2018) [1]**
  congestion from one still frame of a fixed camera
- **Kumar & Raubal (2021) [2]**
  a whole survey of this space — so it is not an opening
- **Feng et al. (2024) [3]**
  detect, track and predict in one pipeline from fixed cameras

None of this gives a controller a per-lane forecast under non-lane-disciplined traffic. That is where the opening is.

SENSOR-GRAPH FORECASTING

- **Yu et al. (2018) [5]**
  graph convolution over space and time
- **Li et al. (2018) [4]**
  diffusion on a directed sensor graph
- **Wu et al. (2019) [7]**
  learns the adjacency instead of taking it
- **Guo et al. (2019) [6]**
  attention over both axes

All four consume loop-detector or probe readings on a known graph. At a camera-only junction those inputs do not exist, so the accuracy is unavailable at any price.

Every mechanism we use is already published

This is stated, not avoided. A reviewer who knows this strand will know it either way.

THE NEAREST NEIGHBOURS

- **Peng et al. (2021) [19]**
  parallel CNN and transformer branches, coupled at every stage
- **Chen et al. (2021) [16]**
  dual branches fused by cross-attention on class tokens
- **Lu et al. (2019) [21]**
  co-attention — bidirectional cross-attention, in 2019
- **Alayrac et al. (2022) [23]**
  learned gates on cross-attention layers
- **Liu et al. (2022) [14]**
  a modern pure CNN matches transformers — the adversarial citation

- **So the architecture is not the contribution**
  the ablation is: does the fusion do anything on this task
- **The gate is treated as an output**
  logged per window and analysed against density — in the works above a gate is an optimisation device reported only through accuracy
- **A gate that learns nothing is reportable**
  and it will be reported

Signal control is solved; anticipation is not tested

WHAT IS SETTLED

- **Wei et al. (2018) [27]**
  deep RL on real signal data
- **Wei et al. (2019) [28]**
  max-pressure reward
- **Wei et al. (2019) [29]**
  network-level cooperation
- **Zheng et al. (2019) [30]**
  phase-competition invariance
- **Chen et al. (2020) [31]**
  decentralised, city scale

Single-intersection control with a policy-gradient method on a simulator is well solved. We claim none of it.

- **Every state vector above is built from the present**
  queue lengths, pressures, occupancies as measured now
- **Whether anticipation helps is open**
  and so is how accurate it must be before it helps
- **That is answerable with a curve**
  no forecast, a noise-calibrated surrogate, an oracle — the sensitivity curve is the contribution on the control side

Viewpoint, not country, is the dataset gap

WHAT EXISTS

- **IDD [34]**
  Indian, heterogeneous — ego-vehicle dashcam
- **UA-DETRAC [35]**
  fixed elevated camera — Chinese traffic
- **CityFlow [33]**
  fixed multi-camera — US traffic

Heterogeneous Indian traffic AND a fixed elevated viewpoint, as continuous video rather than stills, is genuinely under-served. That is a dataset gap, not a modelling one.

- **A moving camera cannot carry per-lane regions**
  they must stay valid across a sixty-step window, so dashcam footage bootstraps detection but cannot supply the corpus
- **So the corpus is curated, not assumed**
  licensed public fixed-camera footage, lane regions placed by hand per camera, labels from PCU-weighted occupancy
- **And this is the project's hardest constraint**
  stated here rather than discovered later

30 papers, compared on the same seven columns

The full table is in the report, on landscape pages. Six rows here show the shape of it.

AUTHOR(S) & YEAR	THEME	METHODOLOGY	LIMITATION / RESEARCH GAP
Chakraborty et al. (2018) [1]	`vision`	Deep CNN on single traffic-camera frames	Scene-level and present-tense, not per-lane
Li et al. (2017) [4]	`graph`	Diffusion on a directed sensor graph	Needs loops and a known road graph
Peng et al. (2021) [19]	`hybrid`	Parallel CNN and transformer, coupled	Nearest neighbour; no gate, no temporal axis
Lu et al. (2019) [21]	`attention`	Co-attention in both directions, 2019	So our fusion mechanism is prior art
Chen et al. (2020) [31]	`control`	Decentralised RL at city scale	State is the present; no anticipation
Varma et al. (2019) [34]	`data`	Indian dataset, heterogeneous classes	Ego-vehicle view cannot hold per-lane regions

WHAT THE LAST COLUMN SAYS

The limitations are not scattered. They concentrate on four assumptions: that sensing infrastructure exists, that a scene-level output is enough, that the present can stand in for the near future, and that traffic is lane-disciplined. Those four are the gap.

The gap, derived rather than asserted

Each claim about what is missing carries the citation of the work that establishes what is present.

SIX LIMITATIONS IN EXISTING WORK

- **The accurate forecasters assume sensing that the target junctions do not have**
- **The camera-based work produces the wrong granularity**
- **Most of it estimates the present rather than the future**
- **Evaluation settings are lane-disciplined**
- **The gate is reported only through its effect**
- **Controller state is built from the present**

- **No method produces the conjunction**
  per-lane, one phase ahead, one fixed camera, heterogeneous traffic, feeding a controller
- **Appearance versus counts is untested here**
  on verified labels with whole cameras held out
- **Nobody reports what a fusion gate learns**
  so its interpretability is unclaimed
- **Forecast-quality sensitivity is unmeasured**
  so a positive control result cannot be attributed to anticipation

What the project does about it

SIX RESPONSES

- **Claim the conjunction, not the components**
- **Build the corpus the gap implies**
- **Make the comparison that decides the question**
- **Treat the gate as a measurement**
- **Answer the control question with a curve, not a comparison**
- **Design for the privacy constraint**

Each response answers a specific limitation from the previous slide. None of them is a general statement of intent.

WHAT HAS NOT BEEN ADEQUATELY ADDRESSED

- **Congestion defined on occupancy rather than count**
- **Per-camera threshold calibration**
- **Camera-level generalisation**
- **Labels that do not derive from the baseline's own inputs**
- **Privacy as a design constraint rather than a policy**

Where Phase I stands

DELIVERED IN THIS REPORT

- **Problem established**
  camera-only, per-lane, one phase ahead, heterogeneous traffic
- **Literature surveyed: 30 papers compared**
  53 references, every one retrieved and cited
- **Gap named and traced to the survey**
  six limitations, five unresolved problems, six responses
- **No measurement reported**
  because none has been made — by construction, not by omission

- **Next: the corpus**
  triage footage, place lane regions per camera, derive PCU-weighted occupancy, label at the sixty-second horizon
- **Then: detection**
  fine-tune for the eight-class Indian taxonomy, per-class accuracy with support counts
- **Then: Phase 1 of the model**
  standard cross-attention to convergence before the gate replaces it — the build order is a constraint, not a preference

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Titles, authors, years, venues and DOIs come from OpenAlex and arXiv, recorded in a committed file. None is reproduced from memory.

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